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Biodesign as an innovative tool to decrease construction induced carbon emissions in the environment

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Abstract: Around 40% of the Earth's carbon emissions are caused by the construction sector, mainly related to materials extraction and energy requirements. Therefore, architects and engineers are looking for solutions to achieve zero carbon emissions with their designs and production methods. This paper aims to investigate innovative construction methods, which are possible by linking the fields of biology and architecture; hence, use them to decrease carbon emissions during construction activities. Although using biomaterials and learning from the examples of and imitating nature is not a new idea, contemporary technological advances and scientific understanding made it possible to emulate more complex ideas from nature. This paper studies three biodesign projects; Neri Oxman's silk pavilion, Wolf Hilbertz's sea-autopia ampere and Biodesign Team Turkey's (TBT) sand architectures to investigate if a low carbon architecture and construction that makes use of biodesign is possible. The innovative construction methods in the cases make use of creatures and natural processes to decrease the amount of raw materials in architectural production, decrease the need for transportation by environmentally friendly onsite production and reduce many steps in the construction process; thus, make it possible to decrease carbon emissions compared to conventional construction methods.

Keywords: bio-architecture; bio-based material; biodesign; eco-construction; sustainability; Neri Oxman; silk pavilion; Biodesign Team Turkey; TBT; sand architectures project; Wolf Hilbertz; sea-autopia ampere; future architecture; bio-calcification; bio-rock; bio-fabrication.

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Biographical notes: Onur Kırdök started working in the field as an Artist Assistant, Field Manager and Project Designer after receiving his Bachelor's degree as an Architect, and co-founded Begonya Mimarlık. His curiosity and enthusiasm for biology and new technologies combined with his passion over regenerative environmentally responsible innovative design and led him to

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Tutku Didem Altun is an Architect and Associate Professor at the Faculty of Architecture, Dokuz Eylül University, Turkey. She is one of the founding members of TBT, she deals with all areas of design, continues to work in the biomimetic, nature-inspired architecture and design-engineering interface. Her main goal is to develop environmental-friendly designs to reduce the damages of the built environment in the world by learning from the miracles of nature.

Deniz Dokgöz is working as an Associate Professor in the Dokuz Eylül University, Faculty of Architecture and in İkiartıbir Architecture Office, of which he is a co-founder. He was awarded 33 prizes from national architectural competitions with İkiartıbir Architecture, among them seven are first and four are equivalent prizes and various buildings, and many of his designs were applied. He served as jury member in various architectural project competitions, and attended various events as guest speaker on competition practice. He was also awarded 20 national and international prizes in the cartoon field.

Ayça Tokuç is an Associate Professor at the Faculty of Architecture, Dokuz Eylül University, Turkey. She is one of the founding members of TBT and has been involved in a number of collaborative projects to enhance the built environment and health in the city. Her research examines sustainable design as both a concept and in implementation in various scales including building and community scales. Her other interests lie in biodesign, regenerative design, innovative design, climate change, energy efficiency and thermal energy storage.

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1 Introduction

Carbon is a key component of all known life on Earth; therefore, carbon in the environment means life. However, the effects of human induced carbon emissions have been widely researched in relation to global warming since the 1990s (Sun, 2017). The impacts of human activities within cities, such as rapid urbanisation or requirement of huge amounts of fossil fuels affect not only cities but also their surrounding areas. IPCC (2018) reports that to reach 1.5°C – the warming limit of the Paris Agreement – carbon emissions should be significantly reduced and cities are the key element in this challenge (Felix and Radhika, 2018). Urban areas are the source of 75% of global CO₂ emissions released into the atmosphere (Koop and van Leeuwen, 2017). The process of construction; from material extraction, to assembly, causes around 40% of the global carbon emissions (Krausmann et al., 2009); therefore, it harms the natural carbon cycle and is considered to have an impact on climate change. Ramesh et al. (2010) reviewed 73 studies related to both residential and commercial buildings from 13 countries, and points out that, 80–90% of global warming potential of life cycle of the building is linked to the operation phase while 10–20% are related to the impacts of construction and

dismantling phase. Another review by Sharma et al. (2011) says that more than 50% of greenhouse gas emissions, in some cases as high as 85%, are caused by the operational phase. Pretot et al. (2014) accounts that the operation phase causes 62–98% of the total effect while construction phase accounts for 1–20% and dismantling phase about 0.2–5%. Even though the amount of impacts from embodied energy seems small, according to Li et al. (2019) they correspond to 0.04 t CO₂-eq/m² in the global scale.

Yeang (2010) points out that designers must consider operational emissions, embodied emissions of construction, and emissions from permanent changes in the amount of onsite vegetation and other sources; yet common methods of construction are generally unsustainable. There are numerous research and experiments for more sustainable construction and urbanisation methods. Scientific and technological developments have provided new tools, materials and more freedom to realise architectural visions. In this context, some designers aim to decrease carbon emissions via utilisation of various technologies such as, passive design strategies (Vale et al., 1991), nature-based building elements (Ottelé et al., 2011), eco-construction and refurbishment (Sodagar et al., 2009), innovative materials (Menges, 2012), reuse of construction waste (Pun and Liu, 2006), nature-based solutions (Kabisch et al., 2016), etc. Some designers aim to cause zero emissions during the operation phase of their buildings and the most common proposals focused on increasing the amount of insulation materials and other energy conservation features. Yet, these buildings still cause emissions during their construction (Kristjansdottir et al., 2016) and the use of additional materials result in an increase in the amount of energy embodied into the building materials (Arrigoni et al., 2017). Chastas et al. (2016) reviewed 90 studies on residential buildings and concluded that the share of embodied energy dominates in low energy buildings (between 26–57%) and nearly zero energy buildings (between 74–100%).

Life cycle analysis studies demonstrate the potential of bio and natural building materials as having low embodied carbon due to their significant carbon storage properties. Since bio-based products can store carbon in approximately 50% of their mass, they can act as a carbon sink. The most common and most studied bio-based material is wood. The carbon calculations for bio-based materials is tricky since the actual amount of stored carbon can be represented by including the amount absorbed before production, however, a common approach is calculating only the carbon absorbed after production as a consequence of material usage (Peñaloza et al., 2016). Pittau et al. (2018) have applied dynamic life cycle analysis to some material configurations including timber joist, straw chips and reed mat; timber frame and hempcrete; and timber frame with conventional insulation materials. They have also analysed common bricks with insulation and concrete with insulation. They found that choice of resource depends on the growth period of bio-based materials, longer rotation periods result in longer time to achieve the stored material to be restored in manufacturing (Pittau et al., 2018).

Material choice does not only effect the energy requirement of the building, raw material extraction and material processing have a significant effect on carbon emissions. Although the use of bio-based materials are an important step to decrease carbon emissions, it does not offer a permanent solution for the future of building technologies. In addition to traditional approaches, multi-disciplinary approaches have opened possibilities in architecture. Deniz and Keskin-Gundogdu (2018) review the use of bacteria, microalgae and fungi for biomaterial production. New building methods with bio-based materials show a great potential, not only because of their carbon capture

capacities, but also lower impact processes related to material extraction and production, transportation, and end of life use (Arrigoni et al. 2017). According to Ingold (2013), design is a process of making that intervenes in already established worldly processes to make the “forms of the living world that we see all around us.” Many new directions opened up in architecture in the 21st century. Some of these offer opportunities for novel low carbon emission construction processes via creating the product by or with the nature itself. Architect Mitchell questions the potential of such a change in the construction process in his Ted speech; “why grow homes?” and replies his own question as “because, we can” (Joachim, 2010). Today, growing a sustainable home with help of the nature and technological developments is a possibility. This aspect is evident in the emerging field of biodesign, where the fields of biology and design come together to solve various problems. The solutions are innovative, have the potential to learn from nature, and reduce the harmful effects of the building industry on the environment.

The needs of the future construction industry cannot be answered with contemporary unsustainable construction materials and methods, new methodologies are necessary. One such method can be biodesign, which makes use of biological growth principles that have surrounded us through nature long before the beginning of our civilisation. This paper aims to investigate novel ways to link biology and architecture via new design and construction tools to decrease construction related carbon emissions in the built environment. Investigating such methodologies to process natural carbon sequestration would balance the current problematic emissions and regenerate the current technologies disrupt and harm to the environment. This hypothesis is discussed within the context of three cases that propose the utilisation of bio-based construction methods; Neri Oxman’s silk pavilion, Wolf Hilbertz’s sea-autopia ampere (Eco-Island City) and TBT’s sand architectures. The objectives include the potential effects of biodesign on decreasing construction induced carbon emissions, specifically through the processes of decreasing the amount of raw material extraction, and utilising innovative construction processes that takes its blueprint from nature and utilises it via physical and biological means.

2 Bio-design through bio-architecture

Nature contains knowledge and experience of limitless possibilities springing from real scenarios that can provide answers to many of today’s questions. Although ‘inspiration from nature’ became a popular concept within the last two decades, humankind has used it in different contexts since antiquity. However, rapid development in biology brought a different dimension to the relationship between nature, technology and design in the second half of the 20th century.

2.1 Bio-architecture

Architects’ duty is to design habitats for our kind, since humanity began to grow in civilisation. Bio-architecture seeks to understand the balance of nature and incorporates it to work from the outset of all parts at various scales. It considers nature as the grandmaster of design, learning from the experience it carries along. Naguib and Hanafi (2013) point out; nature has the ‘most elegant and sophisticated forms’ in terms of engineering principles, which brings out the complicated design principles that life bears in organic structures. With the advent of digital techniques today, we have new

possibilities to push the limits of architecture and engineering; deriving the intellect behind the nature's ingenuity by 'representing, analysing, creating, fabricating, and simulating architectural forms inspired by nature'. It would open a vault of new understanding and evolve the construction process from its roots through the sky by "considerations of aesthetics, materials, structure, and environmental controls" (Naguib and Hanafi 2013).

As Ripley and Bhushan (2016) describe bio-architecture "is not simply copying natural shapes or forms in plan, section and elevation or in ornamentation (on the structure, components or in discrete art objects). This is not simply incorporating bioinspired materials in the construction of a structure or the making of art. It is also not simply scaling up natural principles. Instead, bio-architecture is a multiscale approach to the incorporation into the work of solutions and opportunities that nature presents in solving universal human problems."

In that manner, if we are going to design our own habitats as a civilisation, we should consider collaborating with nature. For a sustainable life in the tomorrow's world that we can live in harmony, bio-architecture carries the potential to grow our civilisation in that quest. According to Naguib and Hanafi (2013), the future world will be designed to function like living organisms, specifically adapted to place and able to draw all of their requirement for energy and water from the surrounding, while the main role of the architect will be how to design 'ecomorphic buildings', buildings that mimic natural systems, which will have the capacity to reconnect people to nature.

2.2 Biodesign

The notion of biodesign is related to bio-architecture. Nature can be called as a designed habitat so that life can happen and continue. In 1997, Janine Benyus's book on biomimicry attracted attention from many disciplines including biology and design. As biologist Benyus (2005) mentioned "Life creates conditions conducive to life. It builds soil; it cleans air; it cleans water; it mixes the cocktail of gases that you and I (we) need to live." In this context, she considers nature as the greatest designer and emphasises the importance of turning to nature and biology to reach these answers. She says that nature has sought for an answer to the vital problems of life for billions of years and achieved results after long trials under various harsh environmental conditions; thus, it actually contains the majority of the answers we seek for our current engineering and design problems; we just need to know where we should look at (Benyus, 1997).

Later, the biomimetic approach, which aims to solve problems by taking inspiration from nature, has become a reference for many disciplines. However, the relations between design and biological sub-disciplines, such as genetic biology, synthetic biology, molecular biology, etc. indicate a different direction for the last 15 years. The concept of biodesign is also often named in relation with biomimicry in the literature yet; the notion of biodesign contains interdisciplinary relations that cannot be explained only by the biomimetic approach and the concept of 'taking inspiration from nature'.

According to the author, curator and instructor Myers (2012), who is one of the well-known authorities that integrate biological processes into architecture and design, biodesign is integrating design with biological systems for achieving better ecological performance. In contrast to bioinspired design, biodesign incorporates living organisms

into design as building blocks, material sources, energy generators, digital storage systems and air purifiers, just to name a few possibilities.

As can be seen from some of the original works of the last ten years, biodesign has gained a new and different dimension. In cooperation with living organisms, biodesign approach makes them a part of the design and the resulting product. It points to a new and different future with the partnership of biology, engineering and design disciplines. Consequently, biodesign is a multidisciplinary approach, which aims to understand and use the mechanism behind the genius of nature, and furthermore, envisions a design practice in collaboration with living organisms to solve design problems.

The concepts of bio-architecture and biodesign show potentials to change the way we think and design. From these points of view, what we call sustainable now is what nature already had before we have disturbed its balance. Nature creates cycles for many things related to life, including carbon, water, raw materials and construction. Since nature creates cycles and is the only carbon neutral example in the world, sustainability in nature is what we should strive for. The following section searches for some answers to how we can learn from nature for a low carbon construction through case studies.

3 Case studies

The three selected case studies represent different approaches to construction. They bring new point of views to today's construction methods, which would be an alternative sustainable solution. Neri Oxman's silk pavilion comes with 3D printing technologies and stitching; Wolf Hilbertz's proposal offers calcification like the coral reefs deep within the ocean water; while TBT's proposal proposes biocalcification in another context.

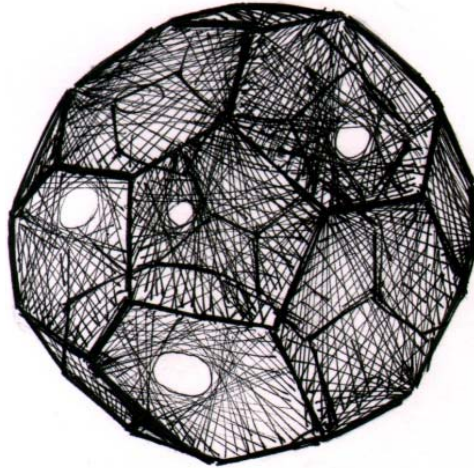
3.1 Neri Oxman's silk pavilion

Neri Oxman and her partners' Silk Pavilion Project (Figure 1), focuses on exploring the possibility of merging digital and biological fabrication technologies within a sustainable design approach. Their answer in this exploration comes from nature, since as Oxman et al. (2017) points out; "most natural and biological materials are made of fibrous structures locally aligned and spatially organised to optimise structural and environmental performance." Nature's own construction processes such as spider nets or bird's nests rely on the animal's ability to generate and work with fibre-based composite materials. For example, spiders' fibres can be optimised for a wide range of conditions including, mechanical properties such as strength and toughness. The final webs have to take into account a delicate balance of function, environmental conditions and material efficiency as limited by the energy resources of a spider.

A silkworm can similarly generate a wide array of mechanical properties related to tensile structures through the silk it produces. To understand the mechanics and what lies behind it, Oxman and partners examined the movements and reactions of *Bombyx mori* silkworm while it is weaving its cocoon in terms of anatomy, behaviour and methods. They used dynamic tracking of silkworms by magnetometer, imaging with wide-angle high resolution micro computed topography and scanning electron microscope, and template fibre-spinning experiments. With the help of these analyses, the logic behind the

movements and features of the silkworm was understood and later transformed into a possible design.

Figure 1 The silk pavilion

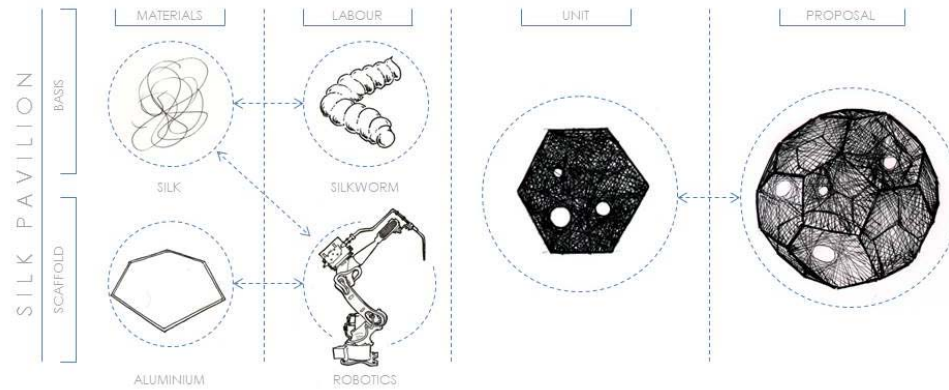


The main inputs in the design phase also featured the behaviour of silkworms against heat and light sources. In addition, a supporting element was considered to control the action of silkworms through the weaving. They used 'RhinoCommon build' that runs 'grasshopper' plug-in to evaluate multiple design solutions within limitations of current computer aided design (CAD). In this context, they designed a panelled dome for the main frame of the pavilion. Its computational projection was examined under solar mapping and aperture distribution mapping. The results provided a multiplier to organise fibre variation. These two maps were used to generate the location, size and density of the pavilion apertures according to rays of natural light entering the pavilion from the south and east elevations, to guide the movement of the silkworms across the structure's surface (Oxman et al., 2014). Consequently, Oxman and colleagues had the chance to direct the design with the cooperation of biological organisms. Before starting with construction, the silkworms' biological characteristics were superposed with these digital mappings to provide maximum silk deposit. This coordinated design method created the opportunity to dictate the 'location, size and density of apertures' and direct the biological organisms.

The construction consisted of a merging between two phases; digital fabrication and biological fabrication. Digital fabrication approach was developed in accordance with a worm's possible range of motion and deposition behaviour. A temporary skeleton of the pavilion was created onto temporary aluminium scaffolding via a computer numerically controlled (CNC) machine that wove silk-based threads as a base. Later, the structure was suspended in space and the frames were removed (Oxman et al., 2017). For the biological fabrication, 6,500 silkworms were fed in a light and temperature controlled room and monitored until they started spinning. Once the worms began spinning, they were transferred onto the structure (Oxman et al., 2017). This spinning process took two to three days and the silkworms that completed spinning were collected on a drop cloth at the bottom of the dome, unlike general silk industry that boils silkworms to take their silk

produce. Oxman et al. (2017) point out that these silkworms are hired to do the labour and “were able to continue their cycle of metamorphosis into a silk moth, including egg laying and reproduction.” The Silk Pavilion Project combines the spinning habits of silkworms as a finishing surface onto a support created by load bearing frames and tensile silk paths, where the labour work was spread between organic and inorganic (Figure 2).

Figure 2 Construction process of the silk pavilion



3.2 Wolf Hilbertz's sea-autopia ampere (Eco-Island City)

Architect and inventor Wolf Hilbertz proposed sea-autopia ampere as a so called autopian dream with the sketches of artist Newton Fallis in the 1970's (Figure 3). It would be "... a marine city that would literally grow out of the sea at Seamount Ampere!" (Hilbertz, 1970; Dobraszczyk, 2017) (Figure 3). The idea behind this proposal comes from an early invention of Hilbertz called 'bio-rock', also known as seacrete or seament. This invention relies on delivering low voltage electrical currents through seawater to submerged metal reef structures via cables. According to Hilbertz (1992), the ocean water covers 70.8% of the surface area of the Earth and is the earth's fluid vein with $1.4 \times 10^9 \text{ km}^3$ of volume. It contains a large mineral reserve of 54.4×10^{15} tons of dissolved salts. It captures around 100.10^9 tons of atmospheric carbon yearly. While the existing worldwide reef area has been estimated in 1992 as $617,000 \text{ km}^2$, coral reefs were estimated to act as a sink for 2.4×10^9 tons of carbon each year (Hilbertz, 1992). Thus, Hilbertz (1992) describes that as direct current passes in seawater, mainly magnesium chloride, CaCO_3 crystals, oxygen and chlorine will form. This would cause dissolved minerals to crystallise onto the structures as a white limestone substrate, similar to what naturally makes up coral reefs. He estimated an expenditure of around 1MW solar energy to produce one ton of building material, of which 0.2238 ton is carbon (Hilbertz, 1992). In addition, this process accelerates the formation and growth of the skeletons of corals and other shell-bearing animals (Porteous, 2009).

Hilbertz's proposal also considers the significant contribution of coral reefs to the ecosystem. Hilbertz explains the significance of coral reefs as; creating an environment with high rates of photosynthesis, nitrogen fixation, and calcium carbonate deposition. They also support larger numbers of plant and animal species than other marine habitats (Hilbertz, 1992). The unique biology of corals played a vital role in determining the reef community's structure, ecology, and nutrient cycle. Yet, these vast reef habitats were and

still are increasingly damaged by bleaching, hurricanes, excessive overfishing, and tourism. The sea-autopia ampere proposal examines the feasibility of repairing and even extending existing reefs. It can be used to stimulate reef and shell-building organisms, creating artificial reef communities and even build a co-existing new habitat for humans.

The sea-autopia ampere project would rise from the shallow waters of Seamount Ampere in the shape of a nautilus (Figure 3). It would use meshing under the sea as a basis. The electricity would be generated and sustained via a photovoltaic array attached on top of the sea. Thus, solar power would give the kickstart for a sustainable construction to grow. In addition, it would support the life around it thus create habitats for the damaged underwater life, while capturing carbon in the environment. Moreover, tidal power can be added to enhance the system, so that the construction will draw nearly all the materials and energy it needs except for the primary materials at the start of the construction or modifications added by its users.

Figure 3 The sea-autopia ampere

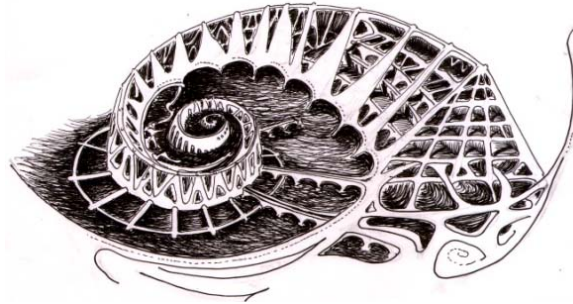
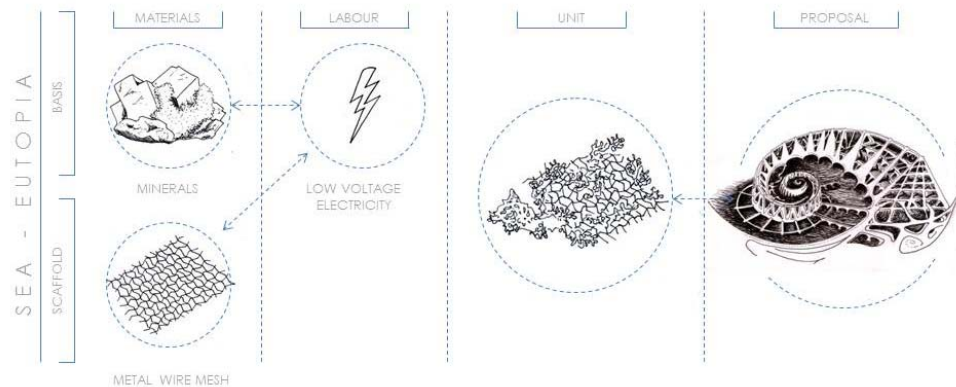


Figure 4 Construction process of the sea-autopia ampere



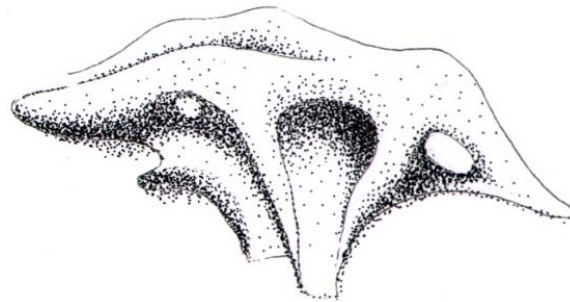
According to Coleman (2007), scientist Thomas Goreau and architect Wolf Hilbertz are the inventors of bio-rock. The group have constructed structures in accordance with the idea of Hilbertz's sea-autopia ampere, as one ongoing step of this project to save the corals in danger. The structures were built in 20 countries including Bali to save the reefs. Goreau has teamed up with many locals and is ready to extend these applications. In principle, the methodology of the applications is the same as sea-autopia ampere. Cables

that carry low-voltage electricity are attached to metal frames and placed in the shores. In addition, broken parts of the coral reefs are salvaged and attached to the structure (Figure 4). Coleman (2007) reports that “Goreau and his supporters say the electricity spurs the weakened coral to revival and greater growth. The electrification program is part of a wider effort in the bay to save the coral.” This method would let the corals grow back from bio-rock structure and to repair the reefs. In this way, bio-diversity of the region can be protected while capturing carbon inside the reefs.

3.3 Sand architectures by TBT

The Sand Architectures Project of Biodesign Team Turkey (TBT) has the theme ‘to leave a sustainable future to our children’ and proposes a children’s playground from sand (Figure 5). It was exhibited in Expo 2016 Antalya, between 23rd of April and 30th of October. The exhibition contained a video and two scaled models showing the concept of the project. The main purpose of the project is to search for alternative methods in the design of urban environments with the help of new scientific developments and technology. In addition to the effects of increasing human population and raw material requirements, desertification is increasing due to climate change. Sand architectures can help in the development of sandy areas in a healthy way, and also has the potential to reduce erosion by holding onto soil and providing an environment for planting (TBT, 2016).

Figure 5 The sand playground

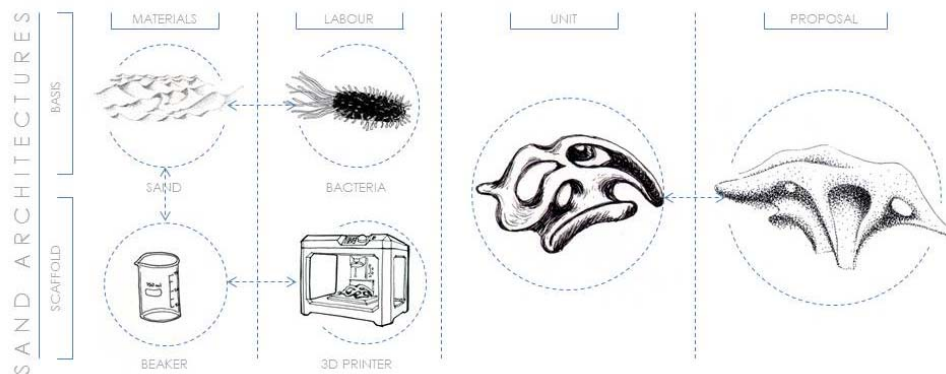


This project focuses on the relationship between biology and design, and in this context, types of bacteria that can produce a structure similar to limestone. The basis of the project is transforming sand into cast on site architectural products by using the urease mechanism to precipitate calcium carbonate with bacteria, such as *Sporosarcina pasteurii* (*Bacillus pasteurii*), which can harden structures like sand or earth. The carbon-dioxide, which the bacteria breathe out, is captured in earth as a part of the calcium carbonate production process. This process already has a wide range of applications, including healing cracks in reinforced concrete buildings, soil stabilisation and environmental processes. TBT has developed a patent-pending technology to give spatial and formal characteristics to the product. Within this framework, different design products can be produced to perform many functions in the design of architectural and urban environments. This playground project was generated as the first example utilising this method. By using sand, bacteria and nutrients as raw materials, a texture quite similar to caves and caverns, which are formed by the water or wind in the nature, is obtained.

The earth is a prominent construction material in nearly all geographies with hot-arid and moderate climates. Nowadays, nearly two thirds of the humans on Earth are living in buildings made of the earth material. Earth construction can have various economic and environmental advantages when compared to conventional building materials and construction methods. Environmental advantages include utilisation of renewable sources, production of less waste, recyclability of most of the waste, less energy requirements, less carbon emissions, non-toxicity and not harming air quality (Pfeifer et al., 2002). The process in sand architectures makes use of all these properties of earth architecture; moreover it requires less resources and energy than traditional methods.

All kinds of urban furniture, architectural/structural/industrial materials/products, such as children playgrounds, seating elements, kiosks, exhibition stands, etc., contain artificially produced materials (concrete, steel, aluminium, plastic, etc.) or processed natural material (wood, stone, etc.). Fjortoft and Sageie (2000) emphasise the significance of simple and natural playgrounds to encourage children's imagination. The coloured playgrounds have negative effects such as inhibiting the physical development of children, and the toxicity of plastic-derived materials. In this context, TBT's proposal can be used to develop architectural/urban/industrial design products that are in harmony with the nature, and suitable to human health with the partnership of bioengineering and architecture disciplines. The sandstone playground architecture emerges after hardening, will be fun and intriguing with its cave-like volume as well as being completely non-toxic for children (TBT, 2016) (Figure 6).

Figure 6 Construction process of the sand playground

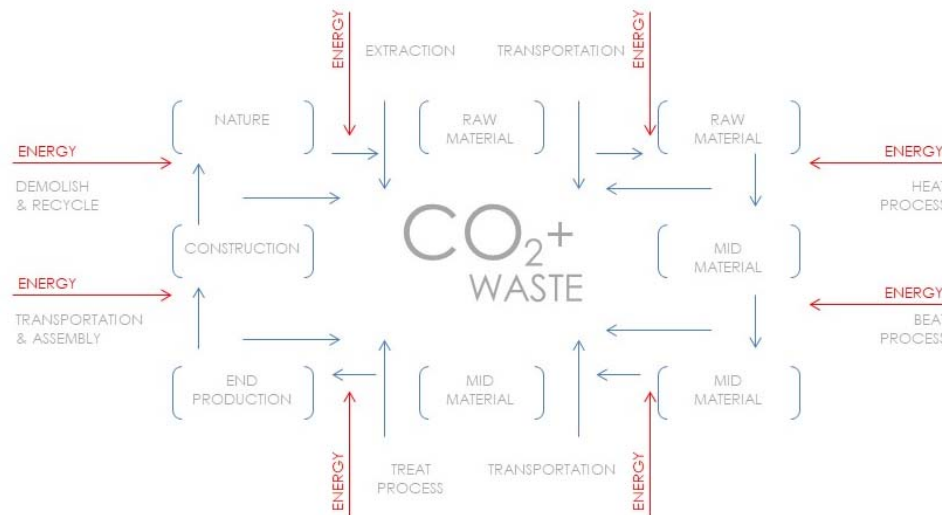


4 Discussion

Today's construction methods are not feasible to bring a more sustainable future. Conventional industry employs the method of 'heat, beat and treat' for production (Benyus, 2005). This methodology requires high temperatures, mechanical and chemical treatments and brings about a great amount of waste from extraction of the raw materials to the finished product (Figure 7). Even green buildings, first release carbon during construction, transportation and production periods, then try to release less carbon during their operation. Fossil fuel use has increased by a factor of twelve during the 21st

century. Meanwhile, raw materials extraction increased by a factor of 34 (COM, 2011). The cases in this study propose to make use of biology and design and provide a different option. The three common aspects behind the case studies are; “how can we decrease the amount of raw materials?”, “how can we decrease the impact of the construction?”, and “how can we provide environmental benefits?” These three aspects are discussed below.

Figure 7 Conventional heat, beat and treat methodology (see online version for colours)



4.1 *Decreasing the amount of raw materials extraction and related carbon emissions*

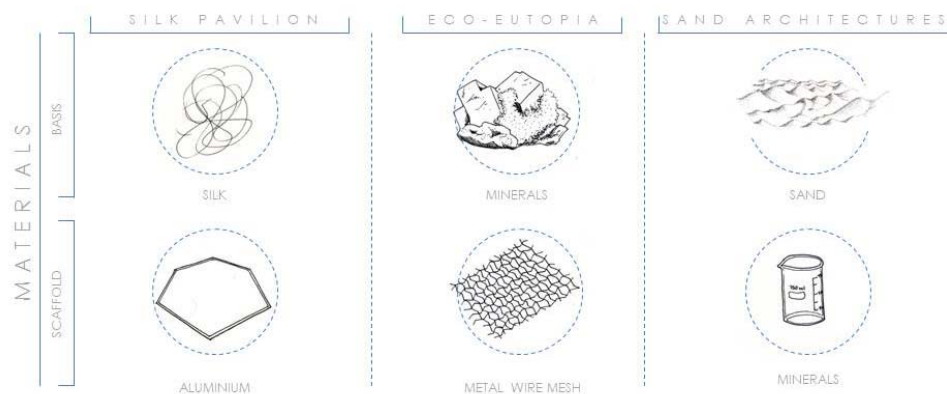
The global construction industry consumes more raw materials than other economic activities, about 3,000 Mt/year or almost 50% by weight (Pacheco-Torgal, 2015). In 2000, the worldwide mining activity generated 6,000 Mt of mine wastes to produce just 900 Mt of raw materials (Whitmore, 2006). In addition to depletion of environmental resources and energy use, environmental impacts caused by this extraction also cause significant environmental threats associated with materials production (Allwood et al., 2011). In this context, the World Business Council for Sustainable Development estimates that materials demand will continue to increase and foresee a necessity to increase resource efficiency four to ten times by 2050 (Pacheco-Torgal, 2015). Yet, the situation in the evaluated cases significantly differs, and can be an answer to this challenge.

The silk pavilion used fibrous elements in a manner analogous to the ingenuity of nature. The principles behind the structure and design were derived from biological organisms. Later, the team led the organisms to produce in a manner that is desirable for them. In addition to using optimum amount of materials in the process, the strength and low weight requirements of the final product were also reached. The sea-autopia ampere uses limestone as a construction material, which has been used since the ancient times.

The technique and materials needed for bio-rock construction is easy to assemble and easy to find even in non-developed countries; thus, this production process of utilising expanded artificial/natural reefs can be used to produce large amounts of a proven building material. While bio-rock requires oceans to grow the material, sand architectures offers a similar of material production that can be shaped and produced on Earth. In addition, this project offers optimum use of materials appropriate to the requirements of the product.

The cases collect their basic ingredients from the nature and emerge on the building site (Figure 8). The utilisation of biologic disposals of living creatures as raw materials changes and shortens the production process. In addition, it decreases the requirements of raw materials, energy, and related carbon emissions. Therefore, the environmental effects in relation to the raw material extraction will decrease. Besides, there will be no heat, beat and treat during the middle steps of the processes; thus the use of high temperatures, toxic chemicals and excessive force will not be necessary. Moreover, optimum material use decreases waste produced during the production process; consequently resulting in less carbon emissions, and may even capture additional carbon as in the case of bio-rock.

Figure 8 Material use in the cases



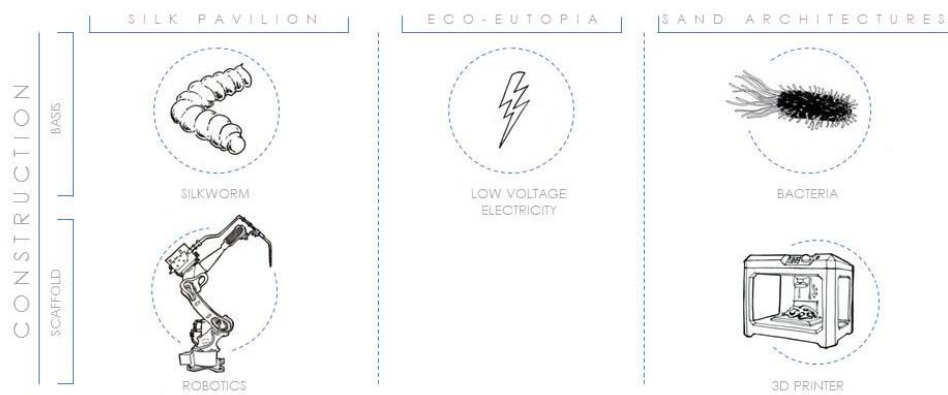
4.2 Decreasing the construction related carbon emissions

Building industry generally deals with end products of considerable mass and size. These constructions either comprise of units installed or crafted/cast on site, and all of the necessary materials have to be transported to the construction site, which cause additional energy and labour for each construction. As discussed earlier, in a conventional construction great amounts of extracted materials and energy are lost before the construction even starts. During the construction, assembly of many parts requires various processes and treatments. Generally, various parts of the construction are mass produced in certain dimensions, therefore, there is almost always a mismatch between what is necessary and the prefabricated parts, which eventually cause more waste.

The cases in this paper bring out a valuable potential solution to this problem; emerging the construction on site. Both Hilbertz's bio-rock and TBT's offer a way to

derive their needed ingredients from the nature. All of sand architectures the bio-rock system works like a ‘sink’ for carbon, thus helps to mitigate global warming, sand architectures is carbon neutral. Oxman’s silk pavilion experiments with a different approach to enhance the mechanics of the construction process – robotics–; and uses the pavilion as a biological swarm for the silkworms. Direct weaving of silkworms onto the construction means that processing silk cocoons into thread and textile will not be necessary. The two systems of 3D printing and biological fabrication complete each other; as the 3D printing constructs the load bearing paths, the biological part fills the structure. In contrast, the biomaterial also forms the structural part in the other cases (Figure 9).

Figure 9 Labour of construction in the cases



Cast on site applications still require transportation of different parts to produce a bigger one; however it eases both logistic problems and the construction process. With onsite construction and more optimum use of materials, the transportation requirements and mid-products of the construction will decrease; thereby the assembly line will shorten and provide more flexibility in design; thus, making mass customisation a possibility. While modularity in a design is another important aspect of the discussed cases, the ability to create a pattern allows the designer to fix or change damaged parts easily. In these cases discussed above, the repair is much more flexible by their nature.

4.3 Other environmental benefits

All of the cases can be used to create awareness in the construction sector, create new green jobs, and a visionary construction sector that rehabilitates instead of destroys nature. As considered in the silk pavilion, use of silkworms promotes a different sustainable approach in finishing fabrication while resulting in a more sustainable silk harvesting cycle than we use today. The use of these construction tools supports life and creates habitats. For example processing of limestone in the sea-autopia ampere proposal will create habitats for the marine life while saving the ecosystem from failing, and increase coastal resilience. The researchers will also educate nearby divers and locals, and offer them a new way of living by involving them in the project. In addition, the

designed products are aimed to be totally natural, non-toxic and robust. The innovative approach of sand architectures via the use of biocalcification and biodegradable biopolymers will be a non-toxic and healthy alternative to the utilisation of plastic elements in playgrounds; thereby enhancing public health and well-being.

5 Conclusions

Even if all the greenhouse gas emissions suddenly ceased, the carbon concentration already in the atmosphere would remain there for the next 100 years and can lead to as much as 3.7°C global warming above pre-industrial temperatures (Pacheco-Torgal, 2015). Meanwhile, the rise in the sea levels, ocean acidification and the occurrence of extreme atmospheric events would continue. As global warming shows its effect, more and more habitats will continue to fail. Yet, the construction sector shows the greatest potential of any single strategy to abate global GHG emissions from the energy sector (IEA, 2012). Therefore, it is necessary to change our view on how we handle construction to mitigate climate change. Some researchers envision that the future of construction will emerge from/by the nature, and nature will be used to grow buildings.

This paper discussed some construction methods that take inspiration from or make use of nature. One of the main advantages of this kind of construction are using fewer raw materials in every step of the production line. In addition, it decreases the production process so that it requires less energy and releases less carbon. Moreover, the process can even capture more carbon than it releases. To realise this kind of design and construction method, a multi-disciplinary team is necessary. However, as we can see in the case of bio-rock, once the technology is developed, the construction of these designs and ideologies can be implemented with contemporary materials, basic tools and unspecialised labour; and help to rehabilitate coral-reef habitats, which have the highest potential to capture carbon. Silk pavilion can be an initial proof-of-concept behind the synthesis of digital fabrication and biological swarm construction, where the robotic arm of a 3D printer and the labour of the silkworm are pioneers of the future of mixed constructions with robotics and biology. All of these cases will emerge on site and shorten the production line, i.e., less carbon emissions released during the construction while using optimum amount of materials.

Although these projects are still in the conceptual phase, they provide clues to the future. The way of rehabilitating the connection we lost with nature is hidden inside the nature. As nature and architecture cross paths today, collaboration between the humanity and nature will lead us to a new path for future of eco-construction. Cooperation with natural fabrication has the potential to become common in the future of the construction industry. All of the discussed cases have the potential to form a basis for new scientific projects and bring a multidisciplinary perspective to the recent tendency towards eco-products or bioproducts. The commercial applications of these emerging bio-architecture products and construction methods can create a new market for production of manmade yet natural environments. While further research is necessary to make these construction tools more commonplace, the construction sector is slowly having such a paradigm shift.

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Might an enormous orbiting "shield" be one way to combat the rising temperatures around the world caused by elevated atmospheric carbon dioxide levels? Writing in the *International Journal of Global Warming*, a team in China has done the calculations and they suggest it might be possible to lower average global temperatures by a third of a degree Celsius with such a shield. This could have a significant, low-cost impact on climate change when we consider that fractions of a degree rises in temperature are already causing serious problems at the Earth's extreme environments [...]

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Biodesign as an innovative tool to decrease construction induced carbon emissions in the environment

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Abstract

Around 40% of the Earth's carbon emissions are caused by the construction sector, mainly related to materials extraction and energy requirements. Therefore, architects and engineers are looking for solutions to achieve zero carbon emissions with their designs and production methods. This paper aims to investigate innovative construction methods, which are possible by linking the fields of biology and architecture; hence, use them to decrease carbon emissions during construction activities. Although using biomaterials and learning from the examples of and imitating nature is not a new idea, contemporary technological advances and scientific understanding made it possible to emulate more complex ideas from nature. This paper studies three biodesign projects; Neri Oxman's silk pavilion, Wolf Hilbertz's sea-autopia ampere and Biodesign Team Turkey's (TBT) sand architectures to investigate if a low carbon architecture and construction that makes use of biodesign is possible. The innovative construction methods in the cases make use of creatures and natural processes to decrease the amount of raw materials in architectural production, decrease the need for transportation by environmentally friendly onsite production and reduce many steps in the construction process; thus, make it possible to decrease carbon emissions compared to conventional construction methods.

Keywords

Author Keywords: bio-architecture; bio-based material; biodesign; eco-construction; sustainability; Neri Oxman; silk pavilion; Biodesign Team Turkey; TBT; sand architectures project; Wolf Hilbertz; sea-autopia ampere; future architecture; bio-calcification; bio-rock; bio-fabrication
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